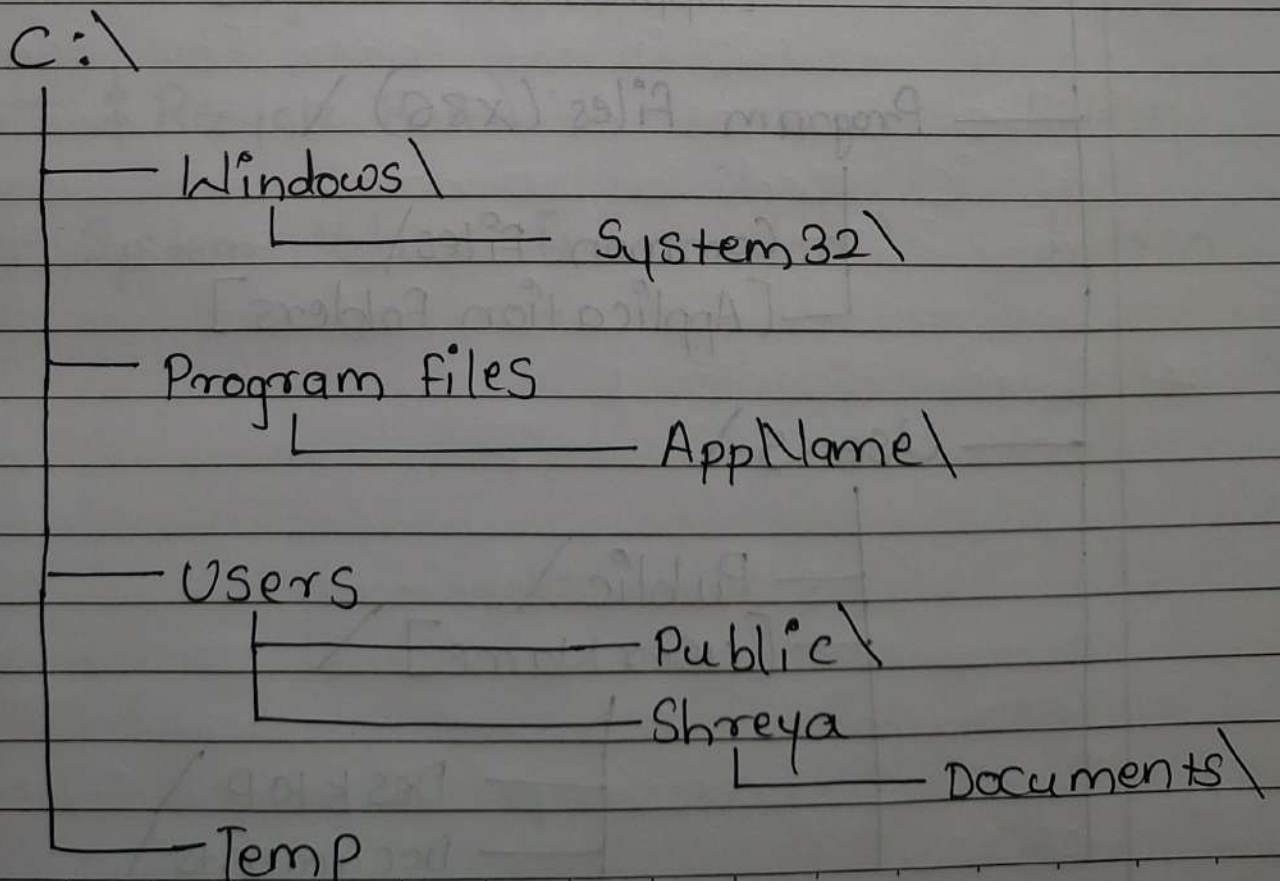


File System

* Windows File System Structure

- The windows hierarchical structure is used to organize and manage files and folders on storage devices.
- it controls how data is stored, named and retrieved by OS
- it organises files and folders in a tree like format, where data is stored at different levels

At the highest level of directory structure we have the root directory, denoted by the backslash character (\). From the root directory we can navigate through different directories and access files and subdirectories.



C:\ (Root directory)

— Windows \

— System32 \

— SysWow64 \

— Boot \

— Fonts \

— Temp \

— Assembly \

— inf \

— WinSxS \

— logs \

— Security \

— Program Files \

— Common Files \

— [Application Folders]

— Program Files (x86) \

— Common Files \

— [Application Folders]

— Users \

— Public \

— [user Name] \

— Desktop \

— Documents \

Downloads \

Pictures \

Videos \

Music \

AppData \

- Local \
- Local Low \
- Roaming \

OneDrive \

Contacts \

ProgramData \

- Microsoft \
- [Application Data]

Perflogs \

\$ Recycle.Bin \ hidden

System Volume Information \ hidden

* C:\ this is the main directory (root)
All other folders and files branch out from here

* Windows\ this is the system directory where all core files needed to run the windows OS are stored.

* Program Files\ This folder holds installed software applications.

* Users\ This directory contains user account data - documents, downloads etc.

* Temp\ This is the folder for temporary files such as drivers DLLs and executables it is used by windows and various applications to store temp. data.

* System Directory
C:\Windows\System32\

System Directory is the critical component of windows operating system. It contains essential system files and libraries necessary for proper functioning of OS

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the OS is defaultly installed in C drive and you can always change the location at your own risk

System32 contains the file that windows need to boot, run and perform everyday tasks. (Engine room of windows)

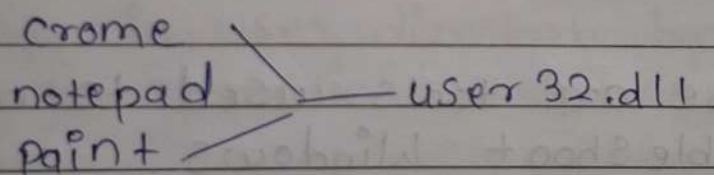
System32\ contains:

1. DLL (dynamic link libraries)

- Code that many program can share
- Instead of every program keeping it's own copy of common functions, windows loads DLL and lets many program use it.

chrome
notepad
paint

user32.dll



```
graph LR; chrome --> user32dll[user32.dll]; notepad --> user32dll; paint --> user32dll;
```

2. EXE files

- these are windows system programs eg. cmd.exe, notepad.exe etc

3. Device Drivers

- Drivers help windows communicate with hardware. eg keyboard, mouse, printer
- without this, windows cannot properly use hardware devices.

4. Configuration Files.

Windows stores Windows Registry hives (databases that contain windows and application settings) these files help windows know:

how to start services

how to load drivers

how system components should behave

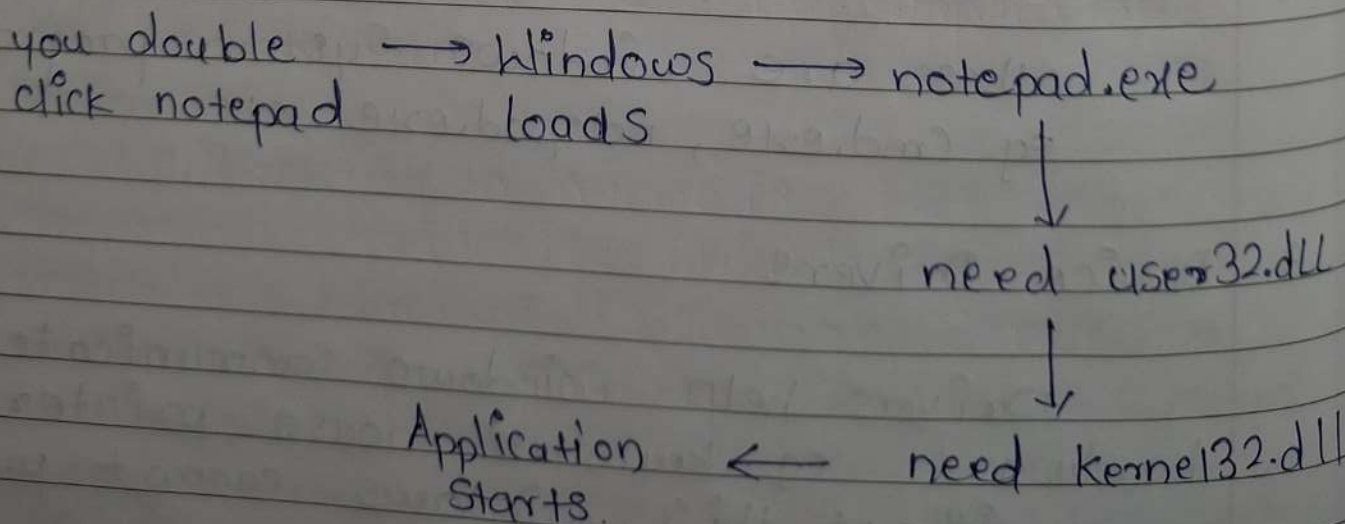
Windows protects these files because they are essential for booting windows & a small change can make windows unable to start so they don't allow to open it.

on Linux, the root user has unrestricted access to almost everything.

5. System Utilities.

These are tools used to manage and troubleshoot windows.

eg.



* C:\Windows\Boot\

it contains files that help windows start (boot).

it is used during the boot process before the windows desktop appears

* User directory

C:\Users\username

the user directory stores personalised settings and files specific to each user account on the system. it contains user-specific data.

In windows 11, microsoft has integrated OneDrive more deeply into the user profile structure, with the OneDrive folder appearing directly in user directory by default and offering automatic backup of Desktop, Documents

* Program Files Directory

program file directory is the default location where applications and programs are installed on the system

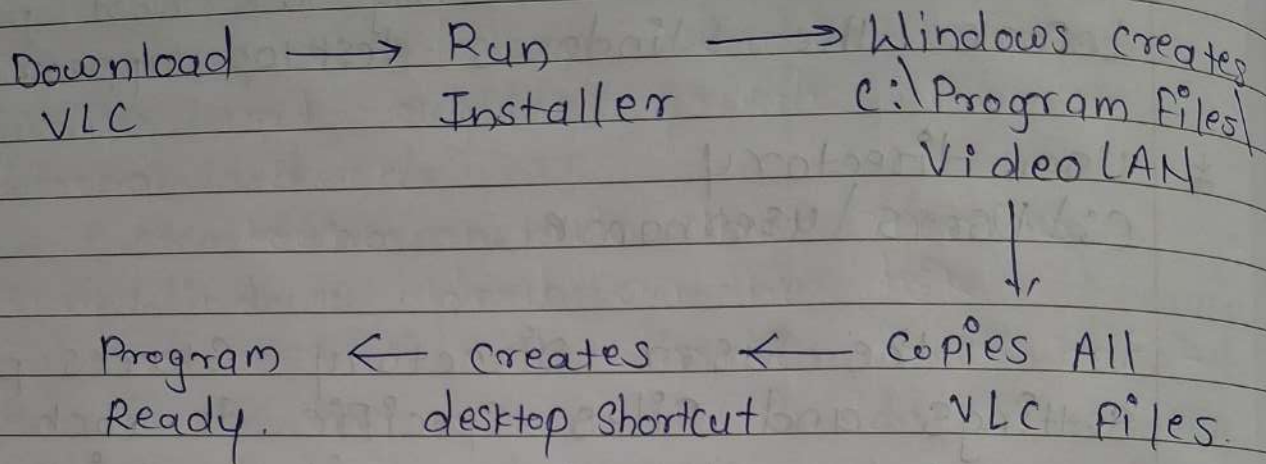
it is divided into two directions

C:\Program Files - this stores 64-bit applications and programs

C:\Program Files(x86) - this stores 32-bit applications and programs on 64-bit system

program files is like home for installed programs

eg. install VLC media player



you should not manually delete a program folder as it will ~~not~~ leave Registry entries behind etc.

* Temporary Files Directory
C:\Windows\Temp

it holds temporary files generated by various processes and applications on the system.

According to Microsoft's best practice, temporary directories should be cleared monthly. Windows 11's Storage Sense can automate this process.

* Program Data Directory C:\Program Data.

it stores application Data that is shared among all users on the computer

Unlike program files, program Data contains variables data like logs, caches, and configurations files that applications need to modify during operations.

C:\Program Data\Microsoft

* WSL Directory Windows subsystem for Linux

WSL lets you ~~test~~ run a linux environment directly inside windows without installing linux as a separate operating system or using a virtual machine.

For WSL, the linux filesystem is stored in a virtual disk file (ext4.vhdx), usually under your user profile.

* Earlier A and B drives were reserved for floppy disks so now following these naming conventions they give name starting from C, D, E to drives.

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* FAT File allocation Table

it is one of the earliest file systems developed by Microsoft for storing and organizing files on storage devices

it keeps a table that records where every file is stored on the disk

it is fast and simple but cannot store files larger than 4 GB. low security

* NTFS new technology File System

It was introduced with windows NT in 1993 and is now the default file system for modern windows installation.

it supports extremely large files and partitions and file compressions.

NTFS maintains a journal that records changes to file system metadata before they are committed.

* HFS hierarchical file system

It is developed by Apple for Macintosh Computers

HFS+ supports much larger sizes than original HFS

* EXT extended File System

the EXT Family was developed for Linux
Features :

Journaling

Large file support

Fast allocation

Better reliability

Delayed allocation

Improved performance

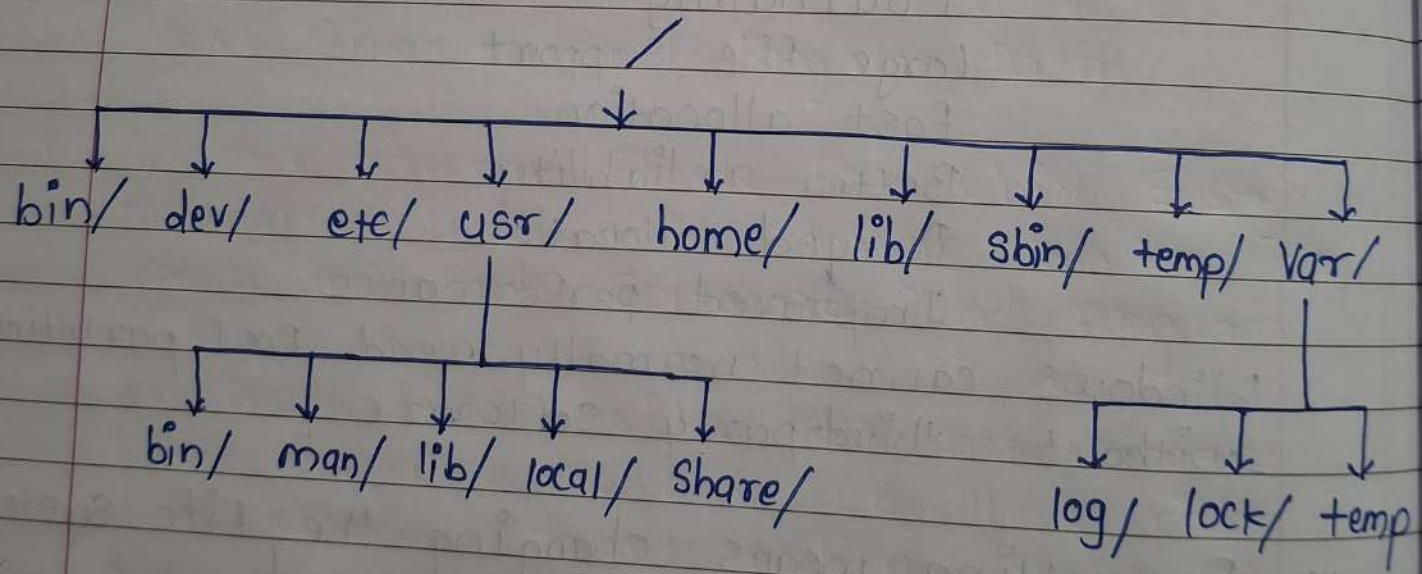
Windows cannot normally read EXT partitions without third-party software.

* Formatting means changing the file system and all existing data is erased

Linux File System Structure.

Linux Foundation maintains a File System Hierarchy Standard (FHS). This FHS defines the directory structure and the content/purpose of the directories in Linux distributions.

Linux is based on UNIX. So you will find same directory structure in BSD and macOS.



* /
the root directory

All the files and directories, in Linux are located under root represented by '/'

* /bin Binaries

it contains the executables files of many basic shell commands like ls, cp, cd
 A binary is a program that has already been compiled into machine code

* /dev Device Files

This directory only contains special files, including those relating to the devices.
 These are virtual files, not physically on disk
 it contains special files that represent the hardware devices connected to your system

/dev/null is a special device that discards everything written to it

/dev/zero produces an endless stream of binary zero bytes whenever it is read.

/dev/random provides random bytes generated by the kernel. Can be used for password or encryption key generation

* /etc Configuration Files

these files tell the operating system and installed programs how they should work. it is like setting folder for linux. Configuration files can store:

Computer hostname

User passwords

Network Settings

DNS settings

Boot Configurations

* /usr User Binaries and Program Data

usr directory stands for Unix System resources. it is main storage location for most installed software and program files.

most files inside /usr are program files. Normal users only use these programs they usually do not modify them so, most of /usr files are read-only for normal users. only administrator can install or modify software there.

* /usr/bin User Commands

this directory contains most executable programs used by normal user

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eg. /usr/bin/python3, /usr/bin/gcc etc

* /bin vs /usr/bin

essential commands

needed during system startup
eg. ls, cp, rm

Most user programs
Installed applications
eg. python, git, java

* /usr/sbin

System administrator commands

This directory contains programs mainly used by system administrator (root)

* /usr/lib

This directory stores libraries used by programs. Library is a collection of ready-made functions that programs can use.

Instead of every program writing the same code again and again, they use shared libraries. It is just like DLL in windows.

* /usr/share

This directory stores files that are shared by all users. They are not executable programs. Instead, they contain documentation, icons, fonts, manuals, language files, help files.

* /usr/src.

Some linux distributions store source code here.

* /home

User personal data.

Home directory contains personal directories for the users. the home directory contains the user data and user-specific config files.

When you create a user on your linux system home directory for user is created eg. /home/Shreya.

* /lib.

Shared libraries.

This directory contains shared library that are required by the essential executable programs stored in /bin and /sbin.

A library is a collection of pre-written functions or code that programs can use.

* /sbin.

This is similar to the /bin directory. the only difference is that it contains the binaries that can only be run by root or a sudo user.

* /tmp Temporary Files

This directory holds temporary files. Many applications use this directory to store temporary files.

* /var Variable data Files

Var, short for variable is where programs store runtime information like system logging, user tracking, caches and other files that system programs create and manage.

the data inside /var keeps changing unlike /bin or /usr, whose contents rarely change, /var is updated continuously.

/var

- log
- cache
- spool
- tmp
- mail
- lib

* /boot Boot Files

This directory contains the files that are required to start the linux operating system.

/boot/vmlinuz is the kernel that manages CPU, Memory, Files, Devices etc

GRUB (GRand Unified Bootloader) is the program that loads the linux kernel.

* /proc process and kernel files

/proc directory contains the information about currently running processes and kernel parameters.

it does not store real files on the hard disk but kernel creates these files in memory (RAM) whenever you access them.

/proc

- cpuinfo
- meminfo
- uptime
- version
- self
- 1
- 2345

} Process ID
PID

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* /opt
Optional directory

Traditionally, the /opt directory is used for installing/storing the files of third-party applications that are not available from the distributions repository.

The normal practice is to keep the software code in opt and then link the binary file in the /bin directory so that all the users can run it.

* /root

root user is the administrator home directory of root user so, instead of /home/root, the home of root is located at /root.

* /media

Mount point for removable media

When you connect a removable media such as USB disk, SD card or DVD, a directory is automatically created under /media directory for them.

* /mnt

Mount directory

This is similar to /media directory but instead of automatically mounting the

the removable media, `mnt` is used by System administrator to manually mount a Filesystem.

*

`/srv`

Service Data

This is the directory where the data is served by services such as a web servers, FTP servers or file servers.

a service is a program that runs in the background and provides some functionality to users or other computers.